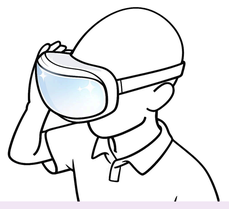
Physical Containers as Framing Conditions for Visualization in Augmented Reality

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Motivation and Research Question

Early-Stage Exploration

Exploratory data analysis (EDA) often starts before users know what to look for [1, 2].

Most visualization systems require users to explicitly configure how data is framed through axes, encodings, or filters.



Why Augmented reality (AR) ?

AR is well suited to this problem because it embeds visualizations in physical environments, allowing framing to emerge from those environments [3, 4].

Research Question: How can **physical containers in AR** shape data interpretation before explicit analytic intent is formed?

Physical Containers as Perceptual Framing Conditions

We reconceptualize **visualization framing in AR** as an environmental condition rather than an explicit user configuration.

- **Container Properties and Framing Tendencies**

Single Face

Single view (overlays for local comparison)

- Landscape rectangle
- Square
- Portrait rectangle
- Circle

Cyclic framing

Multi Face

Planar

Juxtaposed views, Comparative framing

- Horizontal
- Vertical

Volumetric

Spatial encodings such as height & depth

- Cylinder
- Cube
- Limited-face

Thematically grouped views

- **Container Size and Granularity**

	Large container	Small container
	Point-level scatter plot	Aggregated heatmap
	Fine temporal binning	Coarse temporal binning
	More categories visible	Only dominant categories visible

Container geometry shapes three framing dimensions: **view structure**, **granularity**, and **interpretation schema**.

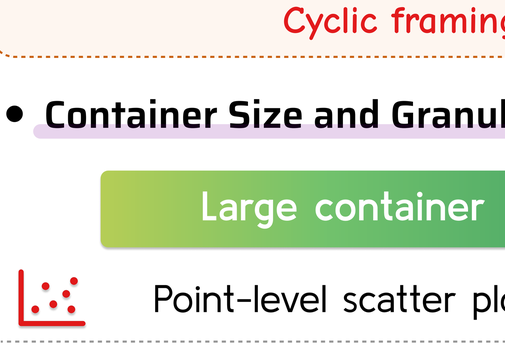
Takeaway

This work extends the unit of analysis beyond the visualization itself to the physical structures that frame it.

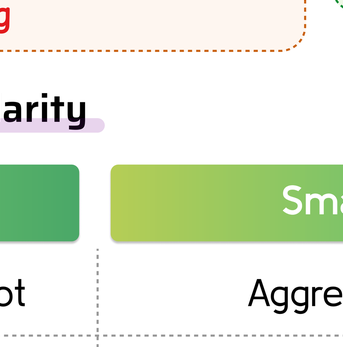
Design Examples

We use a single dataset of monthly movie release counts over three decades.

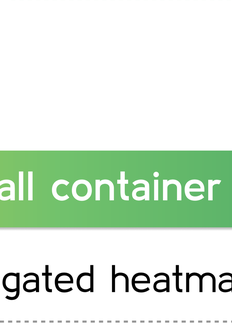
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4



❶ **Single-Face • Rectangle • Overlaid Line Charts**
Enables at-a-glance comparison across decades.

❷ **Single-Face • Circle • Polar Line Chart**
Foregrounds cyclic structure, highlighting seasonal patterns and relative peak months.

❸ **Multi-Face • Planar • Juxtaposed Line Charts**
Highlights within-series temporal structure, including peaks, troughs, and local variation.

❹ **Multi-Face • Volumetric • Helical Time-Series**
Separates seasonal cycles from long-term growth, revealing recurring monthly patterns as a sequential temporal flow.

Each example activates a different perceptual framing tendency.

Limitations

- The taxonomy does not cover all possible container types
- Composite and asymmetric objects remain unexplored
- Warped surfaces may introduce representational distortion

Future Work

- Extend the framework to more complex objects and framing tendencies
- Empirically validate the proposed framing tendencies
- Disentangle geometry effects from encoding effects

Read More

Reference

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[3] M. Whitlock, S. Smart, and D. A. Szafir, “Graphical Perception for Immersive Analytics”, IEEE VR 2020.
[4] N. Elmqvist, “Data Analytics Anywhere and Everywhere,” Communications of the ACM, 2023.